SGA 2: Local Cohomology and Lefschetz Theorems

Exposé VIII: The Finiteness Theorem

Bounded source-aligned working unit — 19 July 2026

Authority. Corrected French lines 2723–2731; original printed pp. 90–91; physical source-PDF p. 80; recomposed running p. 72. Line 2723 resets the equation counter and is included before the proof. Blank line 2732 is excluded; the next source cursor is line 2733. The French and printed proof contain two linked index defects; this target applies and discloses transparent editorial emendations. The French authority remains unchanged, and its compiled reader is not an independent original witness.

Indeed, let $x \in X$, and let U be an affine open neighborhood of x. Let $L^\bullet$ be a projective resolution of the module $\mathcal{F}(U)$, where the L^i are finitely generated. By hypothesis, the ring $\mathcal{O}_{X,x}$ is regular, so the projective dimension of $\mathcal{F}_x$ is finite; denote this integer by d. Set

$$K = \ker(L^{-d} \longrightarrow L^{-d+1}).$$

The module K_x is free, since d is the projective dimension of $\mathcal{F}_x$ ([M], Ch. VI, Prop. 2.1). By (EGA 0_{I}, 5.4.1 Errata), it follows that the $\mathcal{O}_U$-module $\widetilde{K}$ is free on a neighborhood U' of x, with $U' \subset U$. Choose $f \in \mathcal{O}_X(U)$ such that $x \in D(f) \subset U'$. We then have a projective resolution of M_f, where $M = \mathcal{F}(U)$:

$$0 \longrightarrow K_f \longrightarrow (L^{-d})_f \longrightarrow \cdots \longrightarrow M_f \longrightarrow 0.$$

Source note. After choosing only f, French line 2729 and the compiled page print $M_{f'}$, whereas the displayed resolution ends in M_f. Moreover, the corrected kernel is a submodule of L^{-d}, but French line 2730 prints $(L^{d-1})_f$. The target uses M_f and $(L^{-d})_f$ as transparent editorial emendations forced by the chosen localization, the kernel definition, and the displayed resolution. These are not restorations from an independent original scan; the French authority remains unchanged pending editorial review.

This proves that the function under consideration is upper semicontinuous. Since X is quasi-compact, the conclusion follows.